

Multi-Theorem Circle Configurations

Answer Key

High-school Geometry

Quick Answers

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|------------|------------------|--------------------|
| 1. 55 | 5. 125.7 | 9. 55.9 |
| 2. 96, 109 | 6. 56 | 10. 4, -3, 6, 37.7 |
| 3. 12.6 | 7. 5, 3, 5, 1, 9 | |
| 4. 9 | 8. 10 | |
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Curriculum

Level: High-school Geometry

HSG-C.B.5 – problems 1, 2, 3, 4, 5, 6, 8, 9

HSG-GPE.A.1–A.3 – problems 7, 10

Difficulty 2–5 of 5 across 18 tagged checks.

Common wrong answers

6. *If they got 146: forgot that an angle inscribed in a semicircle is a right angle*

1. Circle O : central $\angle AOB = 110^\circ$, C on the major arc. Find $m\angle ACB$.

Solution: $\angle ACB$ is inscribed in the circle and $\angle AOB$ is the central angle standing on the *same* arc AB , so the inscribed-angle theorem applies: the inscribed angle is half the central angle. $m\angle ACB = \frac{1}{2}(110^\circ)$.

$m\angle ACB = 55^\circ$

2. Cyclic quadrilateral $ABCD$ with $m\angle A = 84^\circ$ and $m\angle B = 71^\circ$. Find $m\angle C$ and $m\angle D$.

Solution: In a quadrilateral inscribed in a circle the opposite angles are supplementary, because each pair of opposite angles is inscribed on the two arcs that together make the whole circle ($\frac{1}{2}(360^\circ) = 180^\circ$). $\angle C$ is opposite $\angle A$, so $m\angle C = 180^\circ - 84^\circ$; $\angle D$ is opposite $\angle B$, so $m\angle D = 180^\circ - 71^\circ$.

$m\angle C = 96^\circ$

Then, for the second pair:

$m\angle D = 109^\circ$

3. Circle O , radius 9 cm, inscribed $\angle ABC = 40^\circ$. Find the length of minor arc AC to the nearest tenth of a centimetre.

Solution: Two theorems, in order. *Inscribed angle:* the central angle on arc AC is twice the inscribed angle, so $m\angle AOC = 2(40^\circ) = 80^\circ$, and the minor arc AC measures 80° . *Arc*

length: an arc of 80° is $\frac{80}{360}$ of the circumference,

$$\text{arc } AC = \frac{80}{360} \cdot 2\pi(9) = \frac{2}{9} \cdot 18\pi = 4\pi \approx 12.566$$

12.6 cm

4. Chords AB and CD meet at interior point P with $AP = 8$, $PB = x$, $CP = 6$, $PD = 12$. Find x .

Solution: The intersecting-chords theorem says the two products of the pieces are equal: $AP \cdot PB = CP \cdot PD$, so

$$8x = 6 \cdot 12 = 72 \implies x = \frac{72}{8}$$

Check: $8(9) = 72$ and $6(12) = 72$ ✓

$x = 9$

5. Flowerbed of radius 10 cm; sector AOB has central angle 144° . Find the sector area to the nearest tenth of a square centimetre.

Solution: A sector of 144° is $\frac{144}{360} = \frac{2}{5}$ of the whole disc, so its area is that fraction of πr^2 :

$$\text{area} = \frac{144}{360} \cdot \pi(10)^2 = \frac{2}{5}(100\pi) = 40\pi \approx 125.664$$

125.7 cm²

6. AB is a diameter of circle O , C is on the circle, and $m\angle ABC = 34^\circ$. Find $m\angle BAC$.

Solution: Two theorems again. *Angle in a semicircle:* $\angle ACB$ is inscribed in a semicircle (it stands on the diameter AB), so $m\angle ACB = 90^\circ$. *Triangle sum:* $m\angle BAC = 180^\circ - 90^\circ - 34^\circ$.

$m\angle BAC = 56^\circ$

7. $x^2 + y^2 - 10x - 6y + 9 = 0$. (a) Centre-radius form, centre and radius. (b) The x -values where the circle crosses the x -axis.

Solution (a): Group and complete the square in each variable. Half of 10 is 5 and half of 6 is 3, so add 5^2 and 3^2 to both sides:

$$\begin{aligned}(x^2 - 10x) + (y^2 - 6y) &= -9 \\(x^2 - 10x + 25) + (y^2 - 6y + 9) &= -9 + 25 + 9 \\(x - 5)^2 + (y - 3)^2 &= 25\end{aligned}$$

The centre is (h, k) read off with the opposite signs, and the radius is $\sqrt{25}$.

centre $(5, 3)$, $r = 5$

Solution (b): A point on the x -axis has $y = 0$, so substitute $y = 0$ into the original equation:

$$x^2 - 10x + 9 = 0 \implies (x - 1)(x - 9) = 0$$

Both crossings are real because the distance from the centre to the x -axis is 3, which is less than the radius 5.

$x = 1$ and $x = 9$

8. Tangent $PT = 12$ from exterior point P ; a secant through P meets the circle at A then B , with $PA = 8$. Find $AB = x$.

Solution: The tangent-secant (power of a point) theorem says $PT^2 = PA \cdot PB$. The whole secant is $PB = PA + AB = 8 + x$, so

$$12^2 = 8(8 + x)$$

$$144 = 64 + 8x$$

$$8x = 80$$

Check: $PB = 8 + 10 = 18$ and $8(18) = 144 = 12^2$ ✓

$$x = 10$$

9. Circle O , radius 14 cm, $\angle AOB = 90^\circ$. Find the area of the circular segment cut off by AB , to the nearest tenth of a square centimetre.

Solution: The segment is what is left of the sector once the triangle is removed, so this needs the sector formula *and* the triangle area. *Sector:* $\frac{90}{360}\pi(14)^2 = \frac{1}{4}(196\pi) = 49\pi \approx 153.938 \text{ cm}^2$. *Triangle:* $\angle AOB$ is a right angle between two radii, so the triangle is right-angled with legs 14 and 14: area $= \frac{1}{2}(14)(14) = 98 \text{ cm}^2$. *Segment:* $153.938 - 98 = 55.938$.

$$55.9 \text{ cm}^2$$

10. Fountain $x^2 + y^2 - 8x + 6y - 11 = 0$ (metres). (a) Centre and radius. (b) Area of a 120° sector. (c) Explain the fraction.

Solution (a): Complete the square: half of 8 is 4, half of 6 is 3,

$$(x^2 - 8x + 16) + (y^2 + 6y + 9) = 11 + 16 + 9 \implies (x - 4)^2 + (y + 3)^2 = 36$$

so the radius is $\sqrt{36}$ and the centre is $(4, -3)$ — note the sign flip on the y term: $(y + 3)^2 = (y - (-3))^2$.

$$\text{centre } (4, -3), \quad r = 6 \text{ m}$$

Solution (b): The watered region is a sector of the disc of radius 6:

$$\text{area} = \frac{120}{360} \cdot \pi(6)^2 = \frac{1}{3}(36\pi) = 12\pi \approx 37.699$$

$$37.7 \text{ m}^2$$

Solution (c) — model answer (open response, review by hand): A full turn about the centre is 360° , and the sprinkler sweeps 120° of it. Sectors with the same radius have areas in the same ratio as their central angles, because a sector of angle θ is exactly what you get by cutting the disc into $\frac{360}{\theta}$ congruent pieces and keeping one. Since $\frac{120}{360} = \frac{1}{3}$, the watered region is one third of the fountain's area — and that is why the formula multiplies πr^2 by $\frac{\theta}{360}$ rather than by θ itself.